

Please check the examination details below before entering your candidate information

Candidate surname						Other names					
Centre Number						Learner Registration Number					

Pearson BTEC Level 3 Nationals Certificate, Extended Certificate, Foundation Diploma, Diploma, Extended Diploma

Monday 10 June 2024

Afternoon (Time: 1 hour 45 minutes)

Paper reference **31769H**

Computing
UNIT 2: Fundamentals of Computer Systems

You do not need any other materials.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*

Information

- The total mark for this paper is 80.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*
- You may use a calculator.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

P75938A

©2024 Pearson Education Ltd.
1/1/1/1




Pearson

Answer ALL questions. Write your answers in the spaces provided.

- 1** Ruben runs a day nursery. The nursery provides childcare for children under the age of six.

Ruben is upgrading the computer systems used in the nursery.

The computer systems will be used for:

- communicating with parents and the wider community
- basic computer literacy education for the children
- administrative tasks.

- (a) Ruben has provided desktop PCs for use by the staff and children.

Give **three** advantages of desktop PCs compared to laptops.

(3)

1

2

3



The staff must keep records of each child's progress. Copies of the records are sent to the children's parents or guardians.

These records include:

- copies of drawings and writing the child has done
 - written reports
 - details of the child's progress against specified learning goals.
- (b) State **three** features of word processing software that would help the staff when creating the records.

(3)

1

2

3



(c) Ruben has chosen to use open source word processing software in the nursery.

Explain **two** possible drawbacks of using open source software to create the records.

(4)

1

.....

.....

.....

2

.....

.....

.....



The nursery is open six days a week (Monday to Saturday).

- (d) **Figure 1** shows the number of children who attend the nursery each day for two weeks. The data is represented as matrices.

Week 1	26	34	Week 2	29	19
	42	27		42	41
	21	26		32	18

Figure 1

Calculate the total daily attendance using matrices.

You must show your working.

(2)



- (e) **Figure 2** shows the total daily attendance for the month of April represented as a matrix.

April	130	170
	166	112
	84	108

Figure 2

The matrix in **Figure 2** must be stored in memory as an array.

Write the matrix as an array using row-major order.

(1)



(f) **Figure 3** shows the data for three other months represented as a two-dimensional array.

January	135	161	158	102	96	103
February	108	140	153	138	115	110
March	121	138	120	109	125	123

Figure 3

Explain **two** reasons why a two-dimensional array is a suitable data structure to store this data.

(4)

1

.....

.....

.....

2

.....

.....

.....



(g) The attendance data goes through data validation functions before it is processed.

Explain why data should be validated.

(3)

(Total for Question 1 = 20 marks)



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

BLANK PAGE
QUESTION 2 BEGINS ON THE NEXT PAGE.



2 A local healthcare charity is upgrading its computer systems.

The charity provides support and advice to elderly people and people with specific medical needs.

The charity employs six nurses who share their time working remotely, working in the charity's office, and visiting and supporting people in their own homes.

The charity intends to use a cloud-based server to support the tasks that staff must carry out.

- (a) State **two** ways that the nurses could connect their devices to the cloud-based server when they are not in the office.

(2)

1

2

- (b) The cloud-based server allows staff to access data held on the server using any device if they have an internet connection.

Explain **two other** reasons why the charity would choose to use a cloud-based server.

(4)

1

2



- (c) In addition to the cloud-based server the charity requires a number of different physical devices (e.g. laptops, tablets).

Explain **two** utility software applications that could be used to protect the devices and data.

(4)

1

.....

.....

.....

2

.....

.....

.....



- (d) Data compression is used when transferring data between the cloud-based server and connected devices.

Describe how data compression could improve data transmission speed.

(3)

- (e) The cloud-based server uses multi-factor authentication to protect user accounts.

Users must first enter their username. They must then also enter:

- their password
- and **either**:
 - a 6-digit code that they receive by text message
 - **or** the answer to a security question.

Complete the table to show the logic for the given scenarios.

(3)

Username	Password	6-digit code	Security question	Access granted?
TRUE	TRUE	TRUE	FALSE	
TRUE	FALSE	FALSE	TRUE	
TRUE	TRUE	FALSE	TRUE	



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

BLANK PAGE
QUESTION 2f BEGINS ON NEXT PAGE



- (f) The cloud service provider is developing a script to automatically generate a username for a new user.

Figure 4 shows a flowchart that represents the logic that will be used to develop the script.

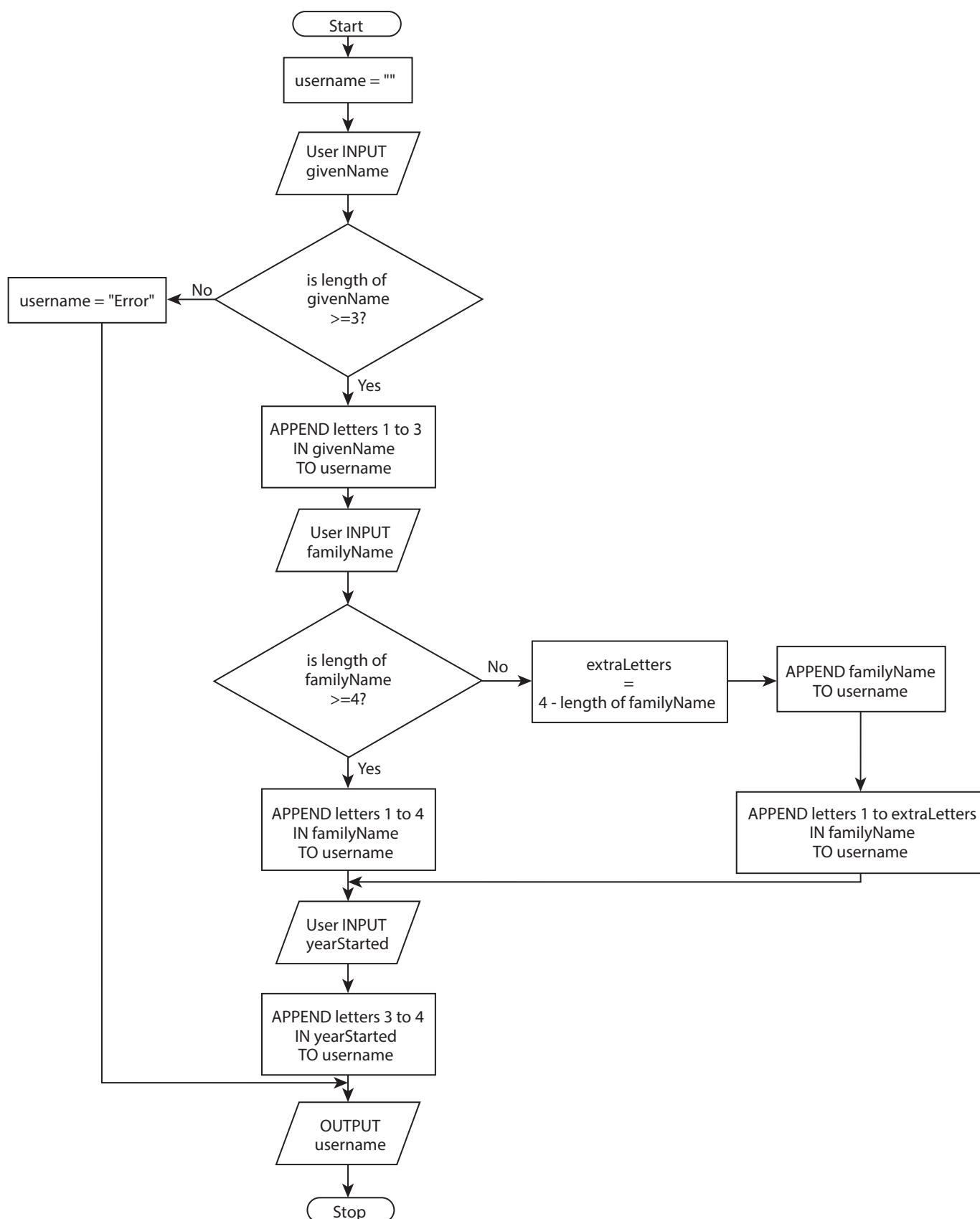


Figure 4



Complete the test table to show what the outputs would be if the script was developed using the logic in **Figure 4**. (4)

Test data	Username output
givenName = Mae familyName = Calderon yearStarted = 2024	
givenName = Jo familyName = Mata yearStarted = 2024	
givenName = Terry familyName = Bowlesworth yearStarted = 2023	
givenName = John familyName = Poe yearStarted = 2023	

(Total for Question 2 = 20 marks)



3 Fran is a computer games designer.

She designs computer games for different platforms and devices including the web, mobile and games consoles.

(a) Fran uses many images in her games.

Describe how reducing the bit depth of an image will impact on the way the image data is stored and displayed.

(4)

(b) Fran uses emulation when designing games.

Discuss the factors Fran will need to consider when using emulation.

You should support your discussion with examples that are relevant to the scenario.

(6)



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Handwriting practice lines consisting of ten horizontal dotted lines.



DO NOT WRITE IN THIS AREA

(8)

[illegible]

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 3 = 18 marks)



- 4** A manufacturing company produces pre-packed meals for a number of different UK supermarket chains.

The company employs over 300 workers including management, admin, production, and technical support staff.

The company uses a range of computer systems for administrative tasks and to control the production process.

It stores and uses a wide range of data including:

- staff personal data
- financial and business data
- production statistics.

- (a) Discuss ways that the company could use computer hardware to protect its data and computer systems.

You should support your discussion with examples that are relevant to the scenario.

(10)

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Handwriting practice area with 20 horizontal dotted lines.



P 7 5 9 3 8 A 0 2 1 2 4

- (b) The company's network manager is assessing the devices used for backup and storage of its data.

The company currently uses:

- solid state drives (SSD) and hard disk drives (HDD) in its servers and devices for storing active data
- external storage media (magnetic tape) to back up data at the end of each day.

Evaluate the suitability of the company's current backup and storage systems.

You should support your evaluation with examples that are relevant to the scenario.

(12)



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 4 = 22 marks)

TOTAL FOR PAPER = 80 MARKS



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

BLANK PAGE

